

NEROLITH

Decision Intelligence Platform

From FloodSim to a Global AI-Powered Geospatial Simulation & Decision Intelligence Platform

Strategic Vision Document · Investor Grade · Confidential

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\$47.8B

\$12.3B

\$6.1B

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Chapter 1

Executive Summary

“The next trillion-dollar opportunity in geospatial intelligence is not better maps. It is better decisions.”

— Nerolith Founding Thesis

Nerolith is an AI-powered Geospatial Simulation and Decision Intelligence Platform that begins as the world’s most capable flood simulation tool and evolves into a full-spectrum, multi-domain simulation layer that sits atop existing GIS infrastructure. We do not compete with ESRI. We make ESRI more powerful.

One-Sentence Pitch

Nerolith is the decision intelligence layer for the physical world — a platform that turns geospatial data into real-time, AI-powered simulation and actionable prediction, for governments, cities, insurers, and infrastructure operators globally.

1.1 The Problem

Today, governments and enterprises possess vast geospatial datasets but lack the tools to simulate future outcomes, test interventions, or make evidence-based decisions under uncertainty. Existing GIS platforms — ESRI ArcGIS, QGIS, MapInfo — are *visualization and data management* tools. They show what *is*, not what *will be*.

Climate change, rapid urbanisation, and infrastructure aging have created an urgent global need for:

- Real-time disaster risk simulation before events occur
- Urban planning tools that model consequences of decisions
- Predictive infrastructure failure analysis

- Multi-scenario decision support for policy makers

1.2 The Solution

Nerolith is built on the foundation of **FloodSim GIS** — a working C++/Qt6 desktop application with a custom ML flood prediction engine, OpenGL 3D rendering, Azure cloud integration, and AI-generated impact reports. This proven technical foundation evolves into a modular, cloud-native platform.

1.3 Key Metrics at a Glance

Target Market	Year-5 ARR Target	Platform Modules
\$47.8B	\$120M	8 Core

1.4 Strategic Pillars

1. **Simulation-First:** Not a GIS clone — a simulation engine that integrates with all major GIS tools
2. **AI Decision Layer:** LLM-powered analysis and recommendations that translate simulation outputs into human decisions
3. **Platform Economics:** API marketplace, developer ecosystem, and partner data network create compounding moats
4. **Vertical Depth** → **Horizontal Scale:** Deep expertise in disaster simulation expands into every domain that needs geospatial decision support

Chapter 2

Current Product Analysis: FloodSim GIS

2.1 What FloodSim GIS Is Today

FloodSim GIS is a professional desktop application built entirely in **C++/Qt6**, targeting disaster management professionals and researchers. It is the only application of its class that combines DEM-based terrain analysis, ML flood prediction, real-time 3D OpenGL rendering, and AI-generated natural language alerts in a single cohesive tool.

Table 2.1: FloodSim GIS Technology Stack

Component	Technology	Role
UI Framework	Qt 6.x (C++)	Widgets, signals, async I/O
3D Rendering	OpenGL 3.3 Core / GLSL	Terrain mesh, shaders, markers
GIS I/O	GDAL Library	GeoTIFF loading, coordinate transforms
Satellite Tiles	ESRI World Imagery	HD satellite texture streaming
Elevation Data	OpenTopography SRTMGL1	30m resolution DEM API
Flood Prediction	Custom ML DLL Engine	Physics + ML flood depth prediction
AI Alerts	Groq API / LLaMA 3.1 8B	Natural language flood warnings
Cloud Storage	Azure Blob Storage (SAS)	Persistent session data
Rainfall Data	Meteostat / RapidAPI	Historical precipitation data
Build System	CMake 3.16+	Cross-platform build configuration

2.2 Core Capabilities

2.2.1 Simulation Engine

The heart of FloodSim is a **custom ML DLL** loaded at runtime, accepting a DEM elevation grid plus rainfall amount and producing a 2D flood depth grid with per-cell risk classification (Low / Medium / High). This architecture — a swappable, runtime-loaded simulation engine with a clean C interface — is the foundational design pattern for the entire Nerolith platform.

2.2.2 3D Terrain Rendering

The OpenGL 3.3 rendering pipeline generates a terrain mesh of 45,000+ vertices with:

- Gaussian-smoothed elevation data for urban terrain fidelity
- Multi-tile ESRI satellite imagery draped over the mesh
- Lambert diffuse + ambient occlusion + atmospheric fog
- Flood depth heatmap layer (Yellow-Orange-Red-Purple gradient)

- Animated billboard risk markers (GLSL pulse animation)

2.2.3 AI Analysis Layer

Two LLM pipelines (Groq LLaMA 3.1 8B) auto-trigger post-simulation:

- **Flood Alert:** Risk level + water depth + recommended action (temperature 0.3 for consistency)
- **Impact Report:** Flood severity distribution, infrastructure stress, mitigation suggestions

2.2.4 2D Analysis Tools

- Marching Squares contour generation with adaptive elevation intervals
- D8 flow direction watershed detection with colour-coded drainage basins
- Real-world flood area calculator using geographic coordinate math
- Meteostat rainfall integration for historically-grounded simulations

2.3 Technical Strengths

Competitive Technical Advantages

- **Runtime-swappable simulation engine (DLL architecture)** — any new simulation type can be added without recompiling the host application
- **Decoupled 2D/3D views sharing a single data layer** — the pattern scales to any number of specialised views
- **Azure-native cloud persistence** from day one — structured run storage is the foundation of a future data platform
- **LLM analysis pipeline** that is domain-grounded (no hallucinated population data) — sets the standard for trustworthy AI in GIS
- **Production-quality OpenGL renderer** — sub-millisecond frame updates, GPU-resident textures, GLSL custom shaders

2.4 Current Limitations

Table 2.2: Identified Gaps and Strategic Response

Gap	Description	<i>Phase Resolved</i>
Desktop-only	No browser/mobile access; Windows 10/11 only	<i>Phase 2</i>
Single simulation type	Flood only; no multi-disaster support	<i>Phase 2</i>
No real-time data feeds	Rainfall is historical, not live sensor data	<i>Phase 2</i>
No collaboration	Single-user, no multi-analyst workflows	<i>Phase 3</i>
No API / SDK	Cannot be embedded in third-party systems	<i>Phase 3</i>
No plugin system	New simulations require core codebase changes	<i>Phase 3</i>
No analytics engine	No dashboards, KPIs, or longitudinal tracking	<i>Phase 2</i>
Single-scenario only	No what-if / scenario comparison	<i>Phase 2</i>

2.5 What Makes FloodSim Genuinely Valuable

FloodSim is not a toy. It is a working, production-quality application that solves a real problem — helping analysts visualise and quantify flood risk against actual terrain, with AI-generated decision support. It reduces hours of expert GIS work to minutes. Most critically, the architecture is *already* designed with platform thinking: swappable engines, separated view layers, cloud-first storage, and AI-native workflows.

Chapter 3

Category Expansion Vision

3.1 Defining the New Category

Nerolith occupies a category that does not yet have a dominant player:

Category Definition

AI-Powered Geospatial Simulation & Decision Intelligence

The layer between raw geospatial data and the human decisions that shape infrastructure, cities, and emergency response — powered by physics simulation, multi-agent modelling, and large language models.

This is not GIS. GIS shows what exists. Nerolith shows what will happen, what could happen, and what should be done.

3.2 Expansion Rationale

The FloodSim core engine is a **terrain-aware, physics-constrained, ML-accelerated spatial simulation system**. The underlying mechanics — DEM loading, spatial grid operations, ML prediction on raster grids, and AI-generated reports — are domain-agnostic. Only the simulation DLL and the domain-specific parameters change across use cases.

3.2.1 Tier 1 Expansion: Multi-Disaster Simulation

The first and most natural expansion reuses 90% of the existing codebase:

Table 3.1: Multi-Disaster Module Map

Simulation Type	Shared Components	New Components
Landslide	DEM, terrain renderer, contours, AI report	Slope stability ML DLL, debris flow model
Wildfire	DEM, risk markers, 2D overlay, AI report	Wind + vegetation fuel map, fire propagation DLL
Urban Flooding	Full FloodSim engine	Storm drain network, impervious surface layer
Pollution Dispersion	DEM, 2D overlay, wind data	Atmospheric diffusion model, source input UI
Earthquake Risk	DEM, terrain renderer, risk markers	Soil liquefaction model, seismic hazard zones
Traffic / Evacuation	Road network overlay, agents	Multi-agent pathfinding engine, capacity modelling

3.2.2 Tier 2 Expansion: Urban Planning and Smart Cities

Once multi-disaster simulation is established, the platform becomes indispensable to urban planners:

- **Development impact assessment:** Simulate how a new building affects local flood risk
- **Green infrastructure planning:** Model the effect of parks, permeable surfaces, retention ponds
- **Climate resilience scoring:** Score every parcel of land against 2050 climate projections
- **Infrastructure stress testing:** Simulate failure cascades across utilities, transport, and communications

3.2.3 Tier 3 Expansion: Predictive Analytics and Decision Support

The final expansion layer converts simulation outputs into forward-looking intelligence products:

- **Risk indices:** Composite, time-series risk scores per geographic unit
- **Policy simulation:** “What happens if we build this levee / rezone this area / increase green cover by 20%?”
- **Insurance risk pricing:** Per-parcel risk models for climate insurance underwriting

- **Supply chain resilience:** Simulate disruptions to logistics networks from climate events

3.3 Why the Expansion Is Logical

The core abstraction is simple: **input a spatial grid + parameters, run a physics/ML model, output a risk/outcome grid, and generate AI-powered recommendations.** This pipeline is identical across all domains. The DEM-based terrain engine, the 2D/3D rendering pipeline, the Azure storage layer, and the LLM analysis chain require zero changes to support new simulation types.

Chapter 4

Platform Architecture

4.1 Architecture Philosophy

Nerolith is architected as a **hexagonal (ports and adapters) system** at platform scale. Each major capability is a discrete module with a well-defined interface. New simulation types, data sources, AI models, and rendering backends are plugins — not core changes. The system is designed so that a third-party developer can build and distribute a simulation module without access to the Nerolith source code.

4.2 Core Platform Modules

4.2.1 Module 1: GIS Data Layer

The GIS Data Layer is the universal data ingestion and normalisation bus. It abstracts all spatial data formats into a common internal representation consumed by every other module.

- **Raster handling:** GeoTIFF, NetCDF, HDF5, COG (Cloud-Optimised GeoTIFF)
- **Vector handling:** GeoJSON, Shapefile, KML, WFS streams
- **Coordinate systems:** EPSG registry integration via PROJ, on-the-fly reprojection
- **Live data connectors:** STAC (SpatioTemporal Asset Catalog) APIs, Copernicus services, USGS endpoints
- **Tile servers:** XYZ/TMS/WMTS compatible; ESRI, Mapbox, OpenStreetMap, custom corporate tile servers

Decoupling mechanism: All data is normalised to a `SpatialGrid<T>` type before leaving this layer. No other module has awareness of the source format.

4.2.2 Module 2: Simulation Engine

The Simulation Engine is a **plugin host**. The existing FloodSim ML DLL is the first plugin. The interface is a clean C ABI:

```
struct SimPlugin {  
    const char* name();  
    const char* version();  
    SimResult run(SpatialGrid* dem, SimParams* params);  
    void free_result(SimResult* r);  
};
```

Engine capabilities:

- **Plugin registry:** Dynamic discovery and loading of simulation DLLs/shared objects
- **Parallel execution:** Multiple simulations run concurrently via thread pool
- **Time-step engine:** Temporal simulation framework (hour-by-hour progression already planned in FloodSim roadmap)
- **Multi-agent system:** Agent framework for evacuation routing, population exposure modelling, infrastructure cascade simulation
- **Scenario manager:** Run N simulations with different parameters, store all results, enable comparison

4.2.3 Module 3: AI Decision Engine

The AI Decision Engine replaces the current single-purpose Groq/LLaMA integration with a full AI reasoning layer:

- **Domain-grounded analysis:** Structured simulation outputs are injected as context — no hallucination risk
- **Recommendation engine:** Given simulation results, the AI proposes mitigation strategies ranked by cost-effectiveness
- **Natural language queries:** Analysts can ask “Which areas would still be safe if the levee on the north river fails?” and receive simulation-grounded answers
- **Automated report generation:** Executive summaries, technical briefings, and public-facing alerts — all from the same data, audience-tailored
- **Model flexibility:** Plugin-style LLM backend — Groq LLaMA, Azure OpenAI, Claude API, local Ollama, or enterprise-hosted models

4.2.4 Module 4: Analytics Engine

The Analytics Engine converts point-in-time simulation results into longitudinal intelligence:

- **Risk dashboards:** Live KPI panels showing risk scores, trends, and anomalies
- **Historical comparison:** Compare current simulation against archived past events
- **Threshold alerting:** User-defined triggers that fire when simulated risk exceeds a threshold
- **Export pipeline:** PDF reports, Excel/CSV data exports, GeoJSON overlays for third-party GIS tools
- **Audit trail:** Full lineage from raw data input to final recommendation — critical for government compliance

4.2.5 Module 5: Rendering & Visualisation Layer

Evolution of the current OpenGL 3D renderer into a multi-target rendering system:

- **Desktop:** OpenGL 3.3+ Core (existing); Vulkan (high-end workstations, Phase 3)
- **Web:** WebGL/WebGPU via Emscripten or separate web renderer (Phase 2)
- **Embedded:** Lightweight raster export pipeline for mobile / low-power devices (Phase 3)
- **Plugin-compatible views:** Any module can register a custom visualisation layer (e.g. a wildfire plugin adds a flame animation layer)

4.2.6 Module 6: Plugin / Extension System

The plugin system is the platform's growth engine:

- **Simulation plugins:** New disaster types packaged as signed DLLs / shared objects
- **Data source plugins:** Custom data connectors for proprietary GIS databases, IoT sensor networks, satellite constellations
- **UI plugins:** Custom panels, tools, and wizards registered by third-party developers
- **Analytics plugins:** New report types, chart types, and KPI definitions
- **Marketplace:** Verified plugin registry with revenue sharing (70% developer / 30% Nerolith)

4.2.7 Module 7: API Layer

The public API layer exposes Nerolith’s capabilities to external systems:

- **REST API:** Trigger simulations, retrieve results, manage projects — full CRUD
- **WebSocket API:** Real-time streaming of time-step simulation progress
- **GraphQL:** Flexible querying of simulation results and spatial data
- **SDK:** Python, JavaScript, and C++ client libraries
- **Webhooks:** Push notifications on simulation completion, threshold breach, or AI alert generation

4.2.8 Module 8: Multi-Tenancy & Collaboration Layer

Enterprise-grade access control and collaboration:

- **Project spaces:** Isolated workspaces per organisation / team
- **Role-based access control:** Analyst, Reviewer, Admin, API-only roles
- **Audit logging:** Every simulation run, export, and API call is logged with user attribution
- **SSO:** SAML 2.0 / OIDC integration for enterprise identity providers
- **Data residency:** Regional cloud deployment (EU, US, APAC) for government data sovereignty

4.3 Module Decoupling Diagram

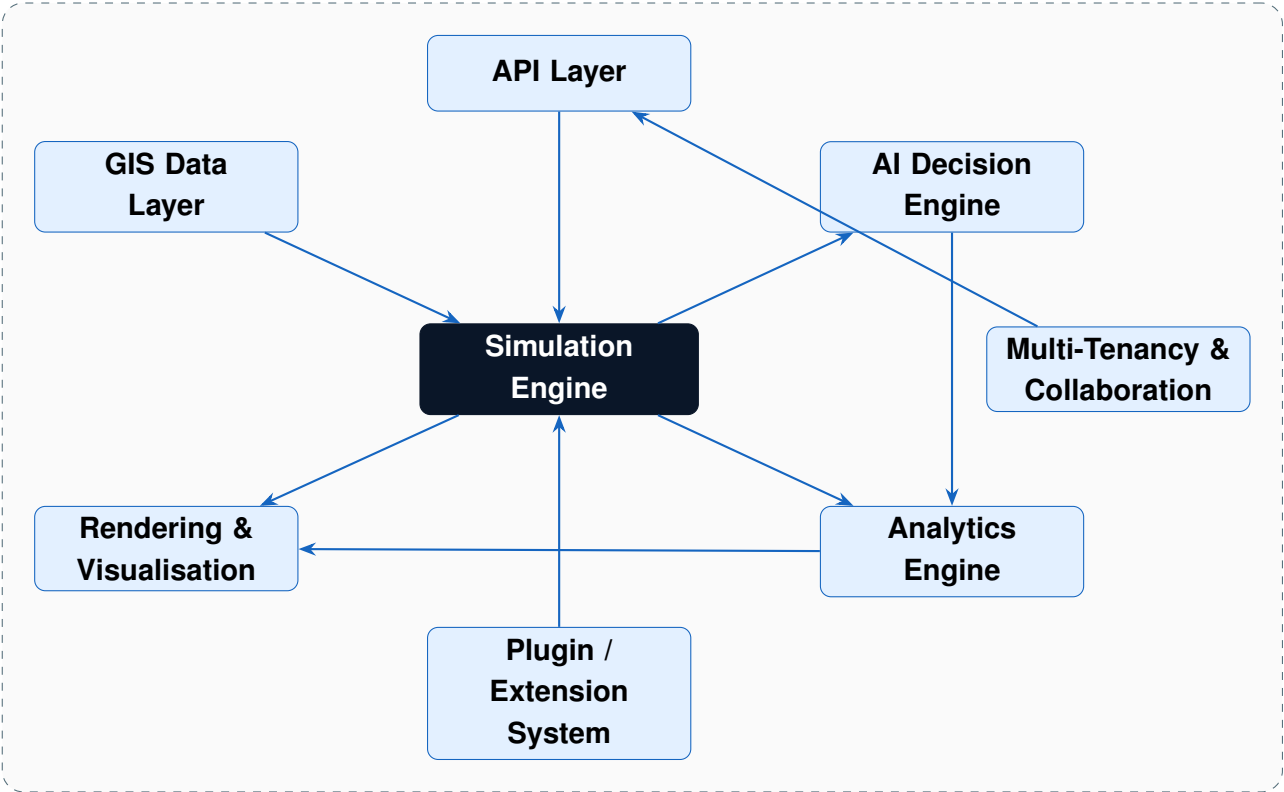


Figure 4.1: Nerolith Platform Module Architecture (all modules communicate via well-defined interfaces)

Chapter 5

Business Model

5.1 Multi-Stage Revenue Architecture

5.1.1 Stage 1 — Niche Product (Year 1–2)

Target: Disaster management agencies, civil engineers, academic researchers, hydrology consultancies.

Product: FloodSim Desktop Pro — the current application, polished and released on the Microsoft Store, with a subscription tier.

Table 5.1: Stage 1 Pricing

Tier	Price	Includes
Community	Free	Core flood simulation, limited cloud storage (1 GB)
Professional	\$49/month	Unlimited simulations, 50 GB Azure storage, PDF reports
Team	\$199/month	5 seats, shared projects, priority support
Academic	\$29/month	Full professional features, educational license

Key metrics target: 2,000 paying users at average \$60/month = \$1.4M ARR by end of Year 2.

Early adopter strategy: Partner with 10–15 national meteorological services and civil protection agencies for pilot deployments. These reference customers are the credibility foundation for government sales.

5.1.2 Stage 2 — Expansion (Year 2–4)

Target: National and regional governments, large municipalities, infrastructure operators (utilities, transport agencies), insurance companies.

Product: Nerolith Multi-Disaster Platform — cloud-native web application with multi-disaster simulation, analytics dashboards, and multi-user collaboration.

Table 5.2: Stage 2 Pricing

Tier	Price	Includes
Government Starter	\$2,500/month	Up to 10 users, 3 simulation types, 500 GB storage
Government Pro	\$8,000/month	Unlimited users, all simulation types, white-label reports
Enterprise	\$25,000+/month	Custom deployment, data residency, SLA guarantees
Insurance Data API	\$0.05/call	Per-parcel risk query API for actuarial workflows

Key metrics target: 80 government/enterprise accounts at average \$6,000/month = \$5.8M ARR by end of Year 4.

5.1.3 Stage 3 — Platform (Year 4–6)

Target: Developers, system integrators, domain experts who build on top of Nerolith.

Product: Nerolith Platform — full API + SDK + Plugin Marketplace.

Table 5.3: Stage 3 Revenue Streams

Stream	Pricing Model	Description
API Access	\$0.02–\$0.50/call	Tiered by simulation complexity and data volume
SDK Licenses	\$5,000/year	Commercial use SDK for embedding Nerolith in third-party apps
Plugin Marketplace	30% revenue share	On all paid plugin sales; 70% goes to plugin developers
Data Subscriptions	\$500–\$5,000/month	Curated geospatial datasets (soil, hydrology, climate projections)
Certification	\$2,000/exam	Nerolith Certified Analyst / Developer certification

Key metrics target: 500 API customers + 50 plugin developers + 200 SDK licenses = \$18M ARR by end of Year 6.

5.1.4 Stage 4 — Enterprise Scale (Year 6–10)

Target: Tier-1 governments (federal agencies, EU institutions, World Bank), global infrastructure operators, sovereign wealth funds investing in climate resilience.

Product: Nerolith Decision Intelligence Suite — a comprehensive platform with digital twin capabilities, real-time sensor integration, and AI-powered policy simulation.

- **National Digital Twin contracts:** Multi-year agreements to model entire nations' risk profiles (\$2M–\$15M per contract)
- **Enterprise Integration:** Connectors for SAP, Salesforce, Microsoft Dynamics, and major SCADA systems
- **Professional Services:** Implementation, training, and custom simulation development (\$300–\$500/hour)
- **Data Network Effects:** As more clients run simulations, Nerolith's aggregated (anonymised) simulation data becomes a proprietary dataset — sold back as benchmarks and industry risk indices

Key metrics target: 15–20 national-scale contracts + broad API economy = \$120M+ ARR by Year 10.

5.2 Revenue Mix Evolution

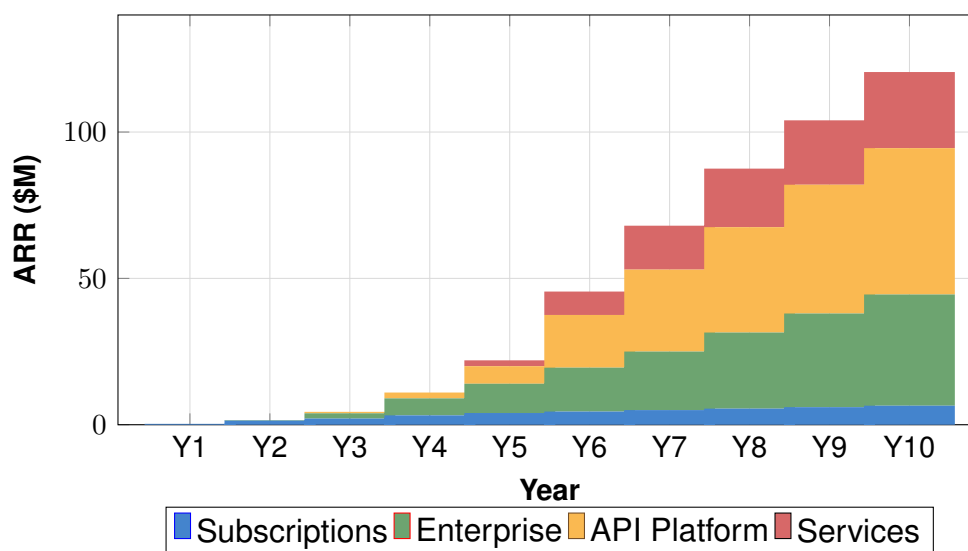


Figure 5.1: Projected ARR by Revenue Stream (Year 1–10)

Chapter 6

Competitive Positioning

6.1 Competitive Landscape

6.1.1 ESRI / ArcGIS

ESRI is the **dominant GIS platform** with over \$2B in annual revenue, serving 90% of the world's GIS professionals. ArcGIS offers extensive data management, cartography, geoprocessing, and basic simulation tools (HEC-RAS integration for hydraulic modelling).

Why Nerolith does not compete with ESRI:

- ESRI sells *data management and visualisation*; Nerolith sells *simulation and decision intelligence*
- ESRI's simulation capabilities are bolt-on integrations (external hydraulic solvers); Nerolith is simulation-native
- ESRI's AI features are generic; Nerolith's AI is grounded in simulation physics
- Most critically: **ESRI is Nerolith's distribution channel**, not its enemy

Strategic relationship: Build an ArcGIS plugin first. Every ESRI user who wants simulation intelligence becomes a Nerolith customer. ESRI's 7M+ active users are Nerolith's top-of-funnel.

6.1.2 QGIS / OSGeo Ecosystem

QGIS is the leading **open-source GIS platform**. It has excellent data management and a large plugin ecosystem but lacks native simulation and AI capabilities.

Opportunity: A free-tier Nerolith QGIS plugin is the most cost-effective developer acquisition strategy. QGIS's 1M+ users — mostly academic researchers, NGOs,

and public-sector GIS teams with limited budgets — are ideal early adopters who will advocate for Nerolith within their organisations.

6.1.3 Simulation-Specific Tools

Table 6.1: Simulation Tool Comparison

Tool	Strength	Weakness	Nerolith Advantage
HEC-RAS	Gold standard hydraulic model	Complex setup, no AI, desktop-only	AI layer + accessibility
MIKE FLOOD	Professional flood modelling	\$50K+ licenses, steep learning curve	10× lower cost, faster setup
InaSAFE (QGIS)	Disaster impact assessment	Limited simulation fidelity	Multi-disaster + live AI
Hazus (FEMA)	US-focused hazard loss estimation	US only, static model, no real-time	Global + dynamic + AI
FloodMap.net	Browser-based flood visualisation	No simulation, visualisation only	Full simulation engine

6.1.4 Emerging AI-Native Competitors

A new class of AI-native geospatial startups (One Concern, Sust Global, Jupiter Intelligence) focuses on climate risk analytics. These companies use ML models trained on historical data. They do not offer physics-based simulation or a general-purpose simulation platform. Nerolith’s hybrid approach (physics + ML + AI reasoning) produces more trustworthy, interpretable results — a critical differentiator in regulated government procurement.

6.2 Differentiation Strategy

The Nerolith Positioning Statement

Nerolith is not a GIS platform. Nerolith is the simulation and decision intelligence layer that makes any GIS platform smarter.

We integrate with ESRI, QGIS, and every major GIS tool. We do not ask users to abandon their existing workflows — we make those workflows dramatically more powerful.

6.2.1 Differentiation Vectors

1. **Simulation-native architecture:** Built for simulation from day one; not retrofitted onto a data management platform
2. **AI that is physically grounded:** LLM outputs are always anchored to simulation results — no hallucination, no generic advice
3. **Accessibility:** Professional-grade results without a PhD in hydraulic engineering
4. **Open plugin ecosystem:** Domain experts can contribute simulation models without Nerolith involvement
5. **Price:** 5–20× lower cost than legacy simulation tools (HEC-RAS consulting + MIKE FLOOD licensing)

Chapter 7

Growth Roadmap: Niche to Platform

7.1 Phase 1 — FloodSim Pro (Months 1–8)

Goal: Ship a polished, monetisable desktop product. Establish credibility.

Key deliverables:

- Time-based flood simulation (hour-by-hour progression with playback controls)
- Evacuation route suggestion using Dijkstra pathfinding on road network overlay
- GeoJSON vector overlay (roads, buildings, critical infrastructure)
- Multi-scenario comparison (run 3 scenarios side-by-side)
- MSIX packaging and Microsoft Store release
- Settings panel replacing environment variable API key management
- PDF report export from within application
- Free community tier + Professional subscription launch

Technical milestone: Refactor FloodEngine DLL interface to the clean C ABI that will become the plugin standard.

Success criteria: 500 active users, 50 paying customers, 3 government pilot agreements signed.

7.2 Phase 2 — Multi-Disaster & Cloud (Months 9–18)

Goal: Become the default simulation tool for disaster management professionals.

Key deliverables:

- Cloud-native web application (React frontend + Python/FastAPI backend)

- Landslide and wildfire simulation modules (first two DLL plugins beyond flood)
- Live data integration: OpenWeatherMap, Copernicus Emergency Management Service
- Analytics engine v1: historical comparison, trend dashboards, threshold alerts
- Multi-user collaboration: shared project spaces, role-based access
- QGIS plugin: free community version drives developer acquisition

Technical transition: Monolithic desktop application splits into:

- **Nerolith Core:** Platform services (simulation engine host, data layer, API)
- **Nerolith Desktop:** Qt6 client (retained for high-performance / offline use)
- **Nerolith Web:** Browser-based interface for collaboration and dashboards

Success criteria: 30 government/enterprise paying accounts, \$3M ARR, 5 simulation module types.

7.3 Phase 3 — API, SDK & Plugin Ecosystem (Months 18–36)

Goal: Become a platform that other products build on.

Key deliverables:

- Public REST + WebSocket API (stable v1.0)
- Python and JavaScript SDKs
- Plugin Marketplace with verification pipeline
- ArcGIS integration plugin (critical: unlocks ESRI's 7M users as distribution channel)
- Developer documentation portal and sandbox environment
- First third-party plugins: earthquake risk (academic partner), pollution dispersion (environmental consultancy)
- Insurance data API: per-parcel flood/fire/landslide risk query endpoint

Technical transition: Full microservices architecture. Simulation engines run as containerised workers (Kubernetes). API gateway handles authentication, rate limiting, and billing.

Success criteria: 200 API customers, 20 plugin developers, 500 SDK downloads/month.

7.4 Phase 4 — Full SaaS Platform (Months 36–60)

Goal: Become the operating system for physical world risk decisions.

Key deliverables:

- Digital Twin infrastructure: persistent, continuously-updated city/region models
- Real-time IoT sensor integration: weather stations, river gauges, seismic sensors
- AI Policy Simulator: “What if” scenario engine with AI-powered consequence analysis
- National-scale deployment capability: handle entire country risk modelling at government grade
- White-label and OEM options for GIS vendors and infrastructure operators
- Nerolith Certification programme: Analyst and Developer tracks
- Advanced analytics: composite risk indices, cross-domain correlation analysis

Success criteria: 5 national-scale contracts signed, \$30M ARR, 500+ API customers.

7.5 Phase 5 — Ecosystem & Data Network (Months 60–120)

Goal: Create compounding moats through data network effects.

Key deliverables:

- Nerolith Benchmark Index: proprietary risk indices derived from aggregated simulation data
- Partner data network: satellite operators, IoT networks, and sensor manufacturers integrate data directly into the platform
- Third-party developer ecosystem: 200+ verified plugins across 15+ simulation domains
- Academic research programme: grants and data access for universities (generates peer-reviewed validation)

- Government data partnerships: national hydrology, geological, and meteorological agencies share authoritative data through Nerolith

7.6 Roadmap Summary Timeline

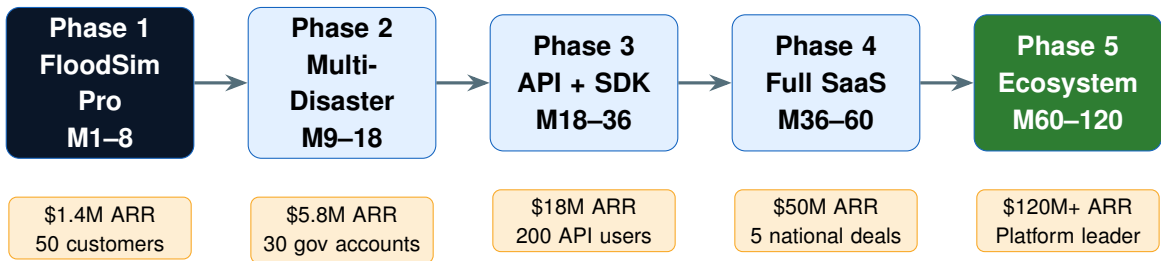


Figure 7.1: Nerolith 10-Year Platform Roadmap with Key Milestones

Chapter 8

Financial & Scaling Strategy

8.1 Unit Economics

Table 8.1: Target Unit Economics by Stage

Metric	Year 2	Year 4	Year 6	Year 10
ARR	\$1.4M	\$9M	\$27M	\$120M
Customer Count	2,000	120	400	1,500+
Avg Contract Value	\$840	\$75K	\$68K	\$80K
Gross Margin	65%	72%	78%	82%
CAC (blended)	\$800	\$12K	\$20K	\$35K
LTV (blended)	\$2,400	\$220K	\$340K	\$640K
LTV/CAC	3×	18×	17×	18×

Note: Customer count decreases from Year 4 to Year 6 in absolute terms because the model shifts from many small subscriptions to fewer, larger enterprise contracts. ARR per customer grows significantly.

8.2 Cost Structure

8.2.1 Engineering

The primary cost driver in Phase 1–3 is engineering talent. The core team requires:

- 2 senior C++/simulation engineers (simulation engine, desktop client)
- 2 full-stack engineers (web platform, API layer)
- 1 ML/AI engineer (flood models, AI decision engine)
- 1 GIS/data engineer (data layer, coordinate systems, data pipelines)

- 1 DevOps/cloud engineer (Azure/AWS infrastructure, Kubernetes)

8.2.2 Infrastructure

Cloud costs are the secondary driver. Nerolith's simulation workloads are compute-intensive but highly bursty — ideal for spot instance pricing on Azure/AWS. Target infrastructure cost: 8–12% of revenue at scale.

8.2.3 Sales and Marketing

Government sales cycles are 6–18 months and relationship-driven. The strategy:

- **Conference presence:** ESRI UC, ICDM, EGU, UN-SPIDER events
- **Academic partnerships:** Free/subsidised access drives peer-reviewed publications that are the most credible marketing in government procurement
- **Partner channel:** Certified implementation partners who sell and deploy Nerolith to their government clients (50% of Year 4+ sales)

8.3 Funding Strategy

Table 8.2: Funding Rounds

Round	Size	Timing	Use of Funds
Pre-Seed / Grants	\$300K	Now	Store release, first government pilots, team of 3
Seed	\$3M	Month 8	Web platform, multi-disaster modules, sales team
Series A	\$15M	Month 24	API platform, marketplace, international expansion
Series B	\$60M	Month 48	National-scale deployments, enterprise sales, ecosystem

Non-dilutive funding targets: EU Horizon Europe (climate resilience), DARPA (disaster simulation), World Bank climate adaptation grants, national science foundations. These programmes fund exactly what Nerolith is building and provide credibility that accelerates government sales.

8.4 Scaling Infrastructure

- **Phase 1:** Azure Blob + Azure Functions; monolithic deployment; <\$500/month

- **Phase 2:** Azure Kubernetes Service; microservices; auto-scaling simulation workers; <\$5K/month
- **Phase 3:** Multi-cloud (Azure primary, AWS/GCP for redundancy); global CDN for tile serving; <\$25K/month
- **Phase 4:** On-premise deployment option for air-gapped government environments; sovereign cloud deployments for data residency compliance

Chapter 9

Unicorn Vision

“The climate crisis has created a trillion-dollar problem that existing software was not built to solve.”

— Nerolith Investment Thesis

9.1 Why This Can Be a Billion-Dollar Company

9.1.1 Market Timing

The convergence of three forces creates an unprecedented window:

1. **Climate urgency:** Governments worldwide are mandating climate risk assessment and resilience planning. The EU Taxonomy Regulation, US SEC climate disclosure rules, and equivalent regulations globally are creating compliance-driven demand for exactly what Nerolith provides.
2. **AI maturity:** LLMs capable of reasoning about domain-specific simulation outputs became available in 2023–2024. The Groq/LLaMA integration in FloodSim is the proof of concept.
3. **Cloud-native GIS adoption:** The shift from desktop GIS to cloud-native platforms is accelerating. ESRI’s ArcGIS Online, Mapbox, and Google Earth Engine demonstrate the market direction. Nerolith is cloud-native from Phase 2.

9.1.2 Moat Creation

Nerolith’s competitive moats compound over time:

Four Compounding Moats

1. **Simulation Data Network:** Every simulation run generates labelled training data. After 100K runs, Nerolith's models are more accurate than any competitor who has not run those simulations. This data moat is impossible to replicate without Nerolith's installed base.
2. **Government Switching Costs:** Government systems integrate deeply. Once a national agency runs disaster simulations on Nerolith, their simulation history, calibrated parameters, and validated models are in the platform. Switching costs are extraordinarily high.
3. **Plugin Ecosystem:** Each domain expert who builds a simulation plugin for Nerolith is simultaneously a customer advocate, a distribution partner, and a barrier to entry for competitors. A wildfire expert who has invested months building a Nerolith plugin will recommend it to every client they serve.
4. **Regulatory Recognition:** Once Nerolith's simulation outputs are cited in government disaster plans and insurance regulatory filings, the platform gains quasi-regulatory status. Competitors must meet the same bar of validation.

9.1.3 Network Effects

- **Data network effect:** More simulation runs → better ML models → more accurate simulations → more customers
- **Developer network effect:** More API customers → more plugins → more use cases → more customers
- **Reference network effect:** Each government customer publishes reports citing Nerolith → legitimises the platform for adjacent governments
- **Data sharing network effect:** Partners who share proprietary data (hydrology agencies, satellite operators) gain access to all other partners' simulation insights — creating a closed data cooperation loop

9.1.4 Market Size Justification

Table 9.1: Total Addressable Market Components

Market Segment	2025 Size	2030 Size	CAGR
GIS Software	\$14.1B	\$22.3B	9.6%
Climate Risk Analytics	\$7.2B	\$12.3B	11.3%
Smart City Software	\$3.8B	\$6.1B	9.9%
Disaster Management IT	\$2.9B	\$5.4B	13.2%
Insurance Climate Tech	\$1.1B	\$4.2B	30.8%
Total TAM	\$29.1B	\$50.3B	—

At \$120M ARR in Year 10, Nerolith captures approximately 0.24% of a \$50B market. This is not a hockey-stick assumption — it is a conservative penetration of a rapidly growing market.

9.1.5 Path to \$1B Valuation

Software-as-a-Service companies with strong revenue growth, high gross margins (>75%), and defensible government contracts trade at 8–15× ARR. At \$120M ARR with 82% gross margins and a platform-level position:

$$\text{Valuation} = 120\text{M ARR} \times 10 \times \text{multiple} = \$1.2\text{B}$$

The multiple expansion pathway: a functional plugin marketplace and data network effects justify SaaS multiples of 15–20×, implying a \$1.8–\$2.4B valuation at the same revenue level.

9.2 The Definitive Statement

Nerolith in Ten Years

The world's national governments, largest cities, and critical infrastructure operators use Nerolith to simulate the future of their physical world — before they build, before disasters strike, and before decisions are made.

Nerolith is not a GIS tool. It is not a simulation tool.

It is the decision intelligence operating system for the physical world.

Appendix A

Glossary of Key Terms

DEM	Digital Elevation Model — a raster grid of elevation values representing terrain surface
GIS	Geographic Information System — software for managing, analysing, and displaying spatial data
GDAL	Geospatial Data Abstraction Library — open-source library for reading/writing raster and vector geospatial data formats
ML DLL	Machine Learning Dynamic Link Library — Nerolith’s runtime-loadable simulation module format
STAC	SpatioTemporal Asset Catalog — open standard for geospatial data catalogue and discovery
LLM	Large Language Model — AI model capable of generating natural language from structured context (e.g. LLaMA 3.1, Claude, GPT-4)
ARR	Annual Recurring Revenue — total predictable revenue expected in a 12-month period from subscriptions
TAM	Total Addressable Market — the total revenue opportunity for a product in its full target market
LTV	Lifetime Value — total revenue expected from a single customer over the entire relationship
CAC	Customer Acquisition Cost — total cost to acquire a new paying customer
Plugin Marketplace	A curated registry of third-party simulation modules, data connectors, and UI extensions for the Nerolith platform
Digital Twin	A continuously-updated virtual model of a real-world geographic area, updated with live sensor data and simulation results

Appendix B

Technical Appendix: FloodSim Module Inventory

Table B.1: Current FloodSim Codebase Summary

File	Module	Key Responsibilities
mainwindow.cpp	Core	Simulation flow, API calls, UI events, Azure upload
mapcanvas2d.cpp	2D View	Satellite tiles, flood overlay, contours, watershed
mapcanvas3d.cpp	3D View	OpenGL init, paintGL, camera, popup overlay
terrainMesh.cpp	3D	Vertex/index buffers, normals, smoothing, skirts
cameracontroller.cpp	3D	View matrix, mouse orbit, pan, zoom
satellitetexture.cpp	3D	Async tile fetch, stitch, QOpenGLTexture upload
riskmarker.cpp	3D	Billboard shaders, pulse animation, marker geometry
vertex.vert	Shader	Transform, UV, normals, world position
fragment.frag	Shader	Satellite, heatmap, lighting, fog, skirt walls
floodengineclient.cpp	Engine	DLL init, loadDEM, predictFlood, isReady